# ANOVA Lecture Activity: Litter Decomposition

## Ecological Context

Litter decomposition is a fundamental ecosystem process affected by microclimate. You'll analyze data from an experiment testing whether leaf litter decomposition differs across canopy types: forest interior, forest edge, and gap. Data of the mass lost from leaf litter in the three canopy types is provided.

# Instructions

1. State your null/alt. hypotheses for ANOVA: Does canopy type affect the amount of leaf mass lost? Make sure your hypotheses are measurable/testable. (1 point)
2. State the mean mass loss associated with each of the three canopy types. (1 point)
3. Conduct a one-way ANOVA.
    a. Go to ANOVA → Classical ANOVA
    b. Set:
        i. Dependent Variable = Mass_Loss_g
        ii. Fixed Factor = Canopy_Type
    c. Check the assumption of homogeneity of variance. Is the assumption met? How do you know? (1 point)
4. Report the f-statistic and the p-value of your ANOVA. What conclusions can you come to regarding your hypotheses (accept/reject)? Be sure to explain what the results of your

ANOVA mean regarding the means of your groups. (2 points)

5. If your ANOVA is significant, perform a Tukey HSD post-hoc test, and report the results, making sure to state what the results mean in terms of group similarities or differences (if it is not significant, state that as well.) (1 point)
6. Create and display a bar plot with error bars (standard error) comparing the three treatments. (1 point)
7. Experimental design: How would you redesign this experiment (or what data would you collect) to test decomposition over time / obtain the decomposition rate? (1-2 sentences) (1 point)

# Deliverables

PDF(s) of questions answered and relevant results from JASP. Feel free to upload any other files that you think may be helpful!

This will be due on Wednesday 10/14 at 11:59PM (after Fall Break!)